IEOR 221

Fall 2025 Final

Due: 5 PM (PST) December 16, 2025

(Please round all your numerical results to 4 decimal places unless indicated otherwise)

1) The current stock price S is 55. It pays 3% continuous dividend yield. The risk free interest rate (continuous compounding) is 4% flat. A European derivative f with payoff

$$Payoff = \max\left(\max(S-50,0), 2(S-75)\right)$$

has 1 year to mature and is traded at 10.94. The implied volatility of a 1 year strike 50 call is 35%.

a) what can you say about the implied volatility surface at T = 1?

b) what is the delta of f ?

2) The current stock price S is 95. It pays annual dividend of $5 in 6 month from now. The current risk free interest rate is 4.25% (continuous compounding). A European style derivative product f_1 with 1 year to mature has payoff

$$f_1 = \begin{cases} 90 - S & S < 100 \\ 0 & S \geq 100 \end{cases}$$

is currently traded at $8.725. What the price of a European style derivative product f_2 with 1 year to mature and payoff

$$f_2 = \begin{cases} S - 90 & S \geq 100 \\ 0 & S < 100 \end{cases}$$

should be?

3) A 2 year zero coupon senior corporate bond with notional N = 1000 is traded 866.9. The risk free interest rate r is 4% (continuous compounding). Assume the bond default recovery rate R is 60%, and assume r, R and default (hazard) rate λ are all constant, what is the fair value of a 2 year junior zero coupon bond with recovery rate R = 20% by the same issuer? The notional N of the junior bond is also 1000. Assume the issuer will not default before the maturity of the bonds.

4) A portfolio consists of two stocks S_A and S_B. Their shares (negative # of shares indicates a short position), current prices, annualized volatility, beta and market consensus (i.e. expectation) for the share prices 1 year from now are given below:

	#shares	price	volatility	beta	expected price in 1y
A	-10000.0	50.0	35.0%	0.250	49.0
B	10000.0	100.0	45.0%	1.500	118.0

The expected return rate of the market is 11%, the market return volatility is 30%, and the risk free rate is 4.0%. Returns of all risky assets follow normal distributions.

a) According to CAPM, what are the fair values for S_A and S_B today?

b) What is the volatility of the portfolio's return?

c) What is the beta of the portfolio?

d) To weather short term market volatility the portfolio manager (PM) decided to make her position market neutral (i.e., beta = 0). The market index future is traded at 8800 per contract. Assuming the beta of the index future = 1.0, how many contracts should the PM long or short?

e) What is the 1-day, 95% confidence level VaR of the portfolio before and after adding the future position? Use 250 as the annualization factor, and assume the index future has the same statistical properties of the market index itself.